

◆ AWS CLOUD PRACTITIONER — SESSION 9

GITHUB LABS

CLOUD MIGRATION CONCEPTS & MODERNIZATION

◆ LAB OVERVIEW

LAB TITLE

Introduction to Cloud Migration & AWS Infrastructure Modernization

LAB DURATION

45–60 Minutes

LAB OBJECTIVES

By the end of this lab students will:

- ✓ Understand traditional vs cloud infrastructure
 - ✓ Deploy a simple cloud environment in AWS
 - ✓ Explore scalability concepts
 - ✓ Understand migration strategies
 - ✓ Create GitHub documentation
 - ✓ Practice cloud modernization thinking
-

◆ REQUIRED TOOLS

Students Must Have

- ✓ AWS Free Tier Account
 - ✓ GitHub Account
 - ✓ Web Browser
 - ✓ Internet Connection
-

◆ GITHUB REPOSITORY STRUCTURE

Students create repository:

Repository Name

cloud-migration-lab-session9

◆ REQUIRED REPOSITORY FILES

```
cloud-migration-lab-session9/
├── README.md
├── migration-notes.md
├── architecture-diagram.png
├── screenshots/
│   ├── ec2-launch.png
│   ├── security-group.png
│   ├── running-instance.png
│   └── aws-console.png
└── reflections.md
```

◆ LAB 1 — CREATE GITHUB REPOSITORY

TASK

Create a new GitHub repository called:

cloud-migration-lab-session9

STUDENT ACTIONS

Step 1

Open GitHub.

Step 2

Create new repository.

Step 3

Set repository visibility:

- Public

Step 4

Initialize with README.

◆ LAB 2 — WRITE CLOUD MIGRATION NOTES

FILE

migration-notes.md

STUDENT TASK

Document:

- what cloud migration is,
 - benefits of migration,
 - traditional infrastructure challenges,
 - migration strategies,
 - modernization concepts.
-

SAMPLE STRUCTURE

```
# Cloud Migration Notes
```

```
## What is Cloud Migration?
```

```
Cloud migration is the process of moving infrastructure and applications  
into cloud environments.
```

```
## Benefits
```

- ```
- scalability
- cost optimization
- automation
- disaster recovery
```

```
Migration Strategies
```

- ```
- lift-and-shift  
- replatforming  
- refactoring  
- hybrid migration
```
-

◆ LAB 3 — LAUNCH AN EC2 INSTANCE

OBJECTIVE

Deploy a basic virtual server in AWS.

◆ STUDENT STEPS

Step 1 — Login to AWS Console

Navigate to:

EC2 Dashboard

Step 2 — Launch Instance

Configure:

- Name: session9-cloud-server
 - AMI: Amazon Linux 2023
 - Instance Type: t2.micro
 - Key Pair: optional
 - Allow HTTP traffic
 - Allow SSH traffic
-

Step 3 — Launch Instance

Click:

Launch Instance

◆ LAB 4 — REVIEW CLOUD BENEFITS

TEACHING OBJECTIVE

Students identify:

- elasticity,
 - scalability,
 - virtualization,
 - automation.
-

DISCUSSION QUESTIONS

Question 1

How is EC2 different from physical servers?

Question 2

What operational costs are reduced in cloud environments?

Question 3

Why is scalability easier in the cloud?

◆ LAB 5 — CREATE SECURITY GROUP

TASK

Review and configure security group rules.

REQUIRED RULES

Type	Port	Source
SSH	22	My IP
HTTP	80	Anywhere

LEARNING OBJECTIVE

Students understand:

- basic network access,
 - cloud security concepts,
 - access control.
-

◆ LAB 6 — DOCUMENT THE MIGRATION SCENARIO

FILE

README.md

◆ REQUIRED CONTENT

Students explain:

Scenario

A traditional company wants to migrate its website into AWS.

Include

Traditional Problems

- downtime,
 - hardware maintenance,
 - scaling limitations.
-

Cloud Benefits

- elasticity,
 - automation,
 - reliability,
 - disaster recovery.
-

SAMPLE README STRUCTURE

```
# Cloud Migration Lab
```

```
## Objective
```

```
Understanding cloud migration concepts using AWS infrastructure.
```

```
## AWS Services Used
```

- Amazon EC2
- Security Groups
- AWS Console

```
## Migration Benefits
```

- scalability
 - operational efficiency
 - reduced infrastructure costs
-

◆ LAB 7 — TAKE SCREENSHOTS

REQUIRED SCREENSHOTS

Students capture:

- ✓ EC2 Dashboard
- ✓ Running Instance

- ✓ Security Group Rules
- ✓ AWS Console Overview

SAVE INTO

/screenshots

◆ LAB 8 — REFLECTION DOCUMENT

FILE

reflections.md

◆ STUDENT TASK

Answer:

1.

What did you learn about cloud migration?

2.

Why do organizations modernize infrastructure?

3.

What advantages does AWS provide?

4.

What challenges exist during migration?

◆ BONUS LAB — ARCHITECTURE DIAGRAM

TASK

Students create a simple architecture diagram showing:

Users → Internet → AWS Cloud → EC2 Instance

TOOLS

Students may use:

- Draw.io
- Canva
- Lucidchart

◆ FINAL DELIVERABLES

Students must push to GitHub:

- ✓ README.md
- ✓ migration-notes.md
- ✓ reflections.md
- ✓ screenshots
- ✓ architecture diagram

◆ GITHUB PUSH COMMANDS

```
git init
git add .
git commit -m "Session 9 Cloud Migration Lab"
git branch -M main
git remote add origin <repository-url>
git push -u origin main
```

◆ INSTRUCTOR REVIEW CHECKLIST

Requirement	Status
Repository Created	☐
EC2 Instance Launched	☐
Notes Completed	☐
Screenshots Uploaded	☐
Reflection Written	☐
Architecture Diagram Added	☐

◆ FINAL LAB MESSAGE

“Cloud migration is the foundation of modern digital transformation and scalable cloud architecture.”